INCH-POUND

MIL-P-46458F(AR)
AMENDMENT 3
March 20, 1990

SUPERSEDING
AMENDMENT 2
27 Sept 1984

MILITARY SPECIFICATION

PROPELLANT M30 FOR USE IN
CARTRIDGE, 105MM, HEAT-T, M456A1
HEAT-T-MP, M456A1E2 AND TP-T, M490

This amendment forms a part of Military Specification MIL-P-46458F (AR) dated 29 January 1980, and is approved for use by the US Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title: Add "(M456A2)" after the designation M456A1E2.

1.1, Line 2: Insert "(M456A2)" after M456A1E2.

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2.1, Drawings: Change "M456A1E2" to "M456A2".

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3.2: Delete in its entirety and substitute:

"3.2 Ballistic assessment requirements. The propellant when fired in the M68 Gun with the M490, M456A1 or M456A2 projectile shall comply with the following requirements at 70 degrees Fahrenheit (degrees F) (see 4.5.1) and standard weapon conditions.

3.2.1 Velocity. The propellant shall be capable of assessment to a velocity of 3850 feet per second (fps).

* 3.2.2: Delete in its entirety and substitute:

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"3.2.2 Pressure. The expected average pressure shall not exceed 59,500 pounds per square inch (psi) for M490, M456A1, nor M456A2."

3.2.3 Charge weight. The calculated charge weight shall not exceed 199 ounces."

3.3, Insert "(M456A2)" after M456A1E2.

3.3.2: Delete in its entirety and substitute:

"3.3.2 Average pressure. At 70 degrees F and standard weapon conditions, the propellant shall not produce an average pressure greater than 59,500 psi for M490, M456A1 nor M456A2. At 125 degrees F standard weapon conditions, the propellant shall not produce an average pressure greater than 68,600 psi when fired with any of the projectiles."

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TABLE I: Delete in its entirety and substitute the following Table I.

* TABLE I: Delete "58700" and substitute "59500".

TABLE I
PROVING GROUND SUMMARY

TEST PHASE	TEMP	SAMPLE SIZE		VELOCITY REQUIREMENTS		PRESSURE REQUIREMENTS	
	Degrees F	Cal	Test	Service Velocity (fps)	Std Dev of Velocity (fps)	IMP (psi)	Average (psi)
CHG ESTAB	70	3	9	3850	INFO	INFO	***
UNIF	-50*	N/A	10	INFO	INFO	INFO	INFO
	70	10	10	3850	7.8	INFO	***
	125	N/A	10	INFO	INFO	73900	**
	145*	N/A	10	INFO	INFO	INFO	INFO

* Performed for initial production testing only.

** The average pressure plus three (3) standard deviations shall not exceed the IMP (see 3.3.3).

*** The maximum average pressure is 58700 for the M490 and M456A1 projectiles and 59500 for the M456A2.

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4.5.1.4.b, Delete everything after 25 fps.

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4.5.4, Insert "(M456A2)" after M456A1E2.

The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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